

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 50

Code of Conduct

I M. Pranay (192525382) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M. Pranay_

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.

A.

Feature	Type-1 Hypervisor	Type-2 Hypervisor
Installation	Runs directly on hardware	Runs on top of a host operating system
Performance	Very high with low overhead	Lower due to host OS overhead
Security	Strong isolation and fewer attack points	Less secure because host OS can be attacked
Manageability	Supports centralized management and live migration	Limited enterprise management
Examples	VMware ESXi, Microsoft Hyper-V, Xen	VMware Workstation, Oracle VirtualBox

Analysis

- Type-1 hypervisors interact directly with hardware, giving faster CPU, memory, and storage access.
- They provide better VM isolation, reducing security risks.
- They support advanced cloud features such as clustering, load balancing, snapshots, and live migration.
- Type-2 hypervisors depend on the host operating system, making them slower and more vulnerable.

Justification

For mission-critical cloud applications, **Type-1 Hypervisors** are the better choice because they provide:

- High performance
- Better security

- High availability
- Efficient resource management
- Enterprise scalability

2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.

A. **Virtualization-based Isolation**

Virtualization enables multiple virtual machines to run independently on a single physical server.

Each workload receives:

- Dedicated virtual CPU
- Virtual memory
- Virtual storage
- Virtual network

This prevents one workload from affecting another.

Isolation Methods

- Hypervisor-based isolation
- Virtual LANs (VLANs)
- Virtual firewalls
- Role-Based Access Control (RBAC)
- Storage isolation

Major Attack Surfaces

1. Hypervisor Attacks

- Malware exploits the hypervisor.
- Can compromise multiple VMs.

2. VM Escape Attack

- An attacker breaks out of one VM and gains access to the host or other VMs.

Conclusion

Proper virtualization architecture ensures secure isolation between finance, healthcare, and student applications while minimizing security risks.

3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.

A.

Analysis

CPU Utilization: 92%

- CPU is heavily overloaded.
- VMs compete for processor time.

Memory Utilization: 88%

- Very high memory usage.
- Host may start swapping memory.

Storage I/O Latency: 35 ms

- High disk latency.
- Slow read/write operations.

Average VM Boot Time: 170 s

- Indicates slow startup caused by CPU and storage bottlenecks.

Likely Bottlenecks

- CPU saturation
- Insufficient RAM
- Slow storage subsystem

- Excessive VM density

Corrective Actions

- Add more CPU cores.
- Increase RAM capacity.
- Upgrade HDD to SSD/NVMe storage.
- Balance workloads across multiple hosts.
- Enable resource monitoring and load balancing.
- Remove unused VMs.

Expected Outcome

- Lower CPU utilization
- Reduced memory pressure
- Faster storage access
- VM boot time below 60 seconds

4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.

A.

Traditional Datacenter

- Manual hardware configuration
- Slow resource provisioning
- Separate management of compute, storage, and networking
- Limited scalability

Software Defined Datacenter

All infrastructure resources are virtualized and managed through software.

Compute Virtualization

- Multiple VMs on one server
- Dynamic resource allocation
- Live migration
- Automatic scaling

Storage Virtualization

- Pools physical storage into one logical storage system
- Thin provisioning
- Faster backup and disaster recovery

Network Virtualization

- Software Defined Networking (SDN)
- Virtual switches and routers
- Automated network configuration
- Improved security through micro-segmentation

Advantages

- Faster deployment
- Better scalability
- Reduced operational cost
- Improved resource utilization
- Easier disaster recovery
- Centralized management

Conclusion

SDDC significantly improves agility by automating compute, storage, and network management while reducing manual intervention.

5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

A.

Problem Analysis

Different cloud providers maintain separate identity systems.

Issues include:

- Different user IDs
- Different authentication methods
- Different access policies
- Poor synchronization

This causes login failures and access control problems.

Secure Identity Design

Identity Federation

- Single Sign-On (SSO)
- Shared identity across cloud providers

Identity Provider (IdP)

- Authenticates users
- Issues secure authentication tokens

Service Provider (SP)

- Trusts the Identity Provider
- Grants access after verifying tokens

Privacy Protection

- Multi-Factor Authentication (MFA)
- End-to-end encryption
- Role-Based Access Control (RBAC)
- Least Privilege Principle
- Audit logging and monitoring

Benefits

- One login across multiple clouds
- Strong security
- Better privacy protection
- Reduced identity mismatches
- Easier user management

Conclusion

Using identity federation with SSO, MFA, RBAC, and encryption provides secure authentication, protects user privacy, and enables seamless access across multiple cloud providers.

Analytical ProblemSolving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded: